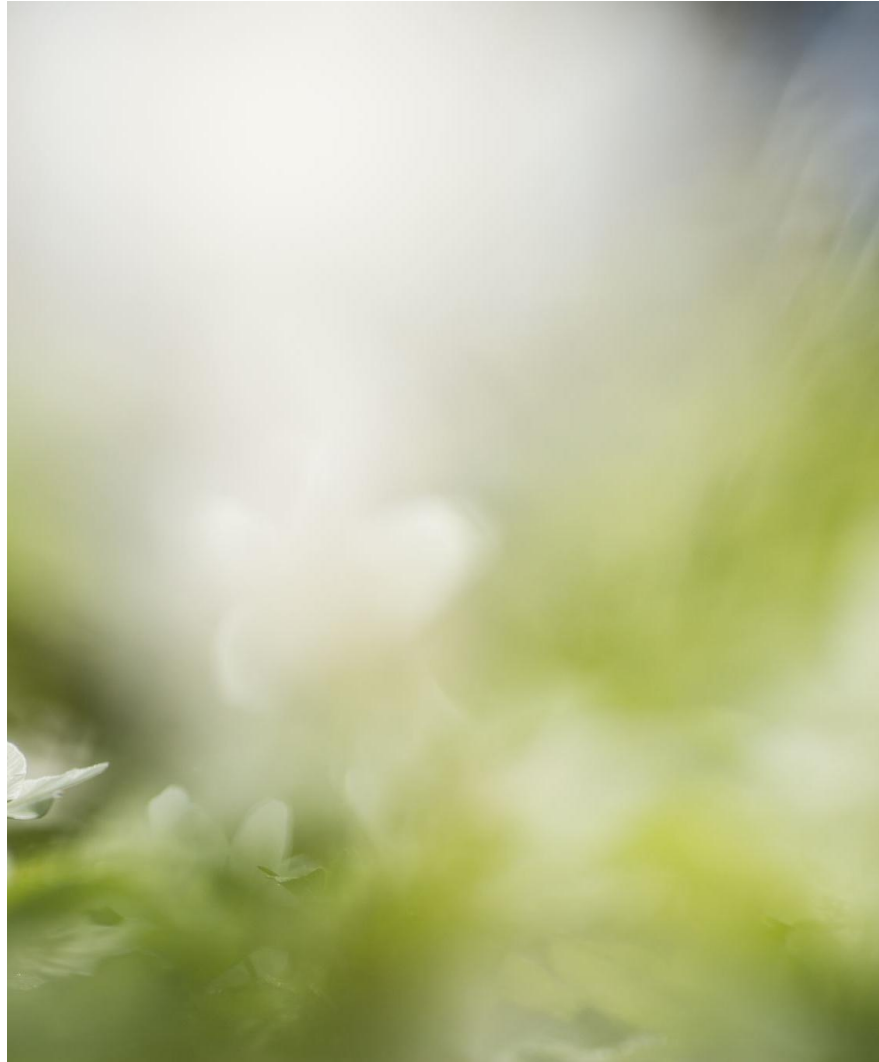




Spring AI

CONNECTING YOUR
ENTERPRISE DATA &
APIS WITH AI
MODELS

1. Generative AI Concept
2. The Importance of Java/Spring in AI
3. Spring AI: A GenAI Integration Solution for Spring Boot
4. Architectural Components and Processing Flow
5. Examples and Demo
6. Notes and Best Practices
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Generative AI Concept

What is Spring AI?

Spring AI is a project that aims to **streamline the development** of applications incorporating Artificial Intelligence functionality within the familiar Spring ecosystem.

Inspired by Python projects like LangChain and LlamaIndex (but not a direct port).

Goal: Bring Generative AI to all programming languages—not just Python.

Key Features

AI that creates new content based on learned data:

- Text: emails, blogs, code, Q&A
- Images: graphic design, text-to-image
- Audio/Video: music, 3D models

Key Features

Core technologies

- Transformers (GPT, BERT)
- LLMs (GPT-4, Claude 3)
- Diffusion Models (DALL·E, Midjourney)
- Embeddings & Vector Space (RAG)
- Prompt Engineering

How GenAI Works – Data & Training

Data collection & preprocessing:

- Diverse sources: text, image, audio
- Cleaned and structured for learning

Model training:

- Deep learning with Transformers
- Predicts next token based on context
- Requires massive compute (GPU/TPU)

How GenAI Works – Data & Training

Large Language Models (LLMs)

- GenAI typically involves extremely large models with billions of parameters.
- Depend on their vast number of parameters and diverse training data, these models can generate new, natural, and varied content.
- Example: GPT-4 can answer questions, write code, and produce creative natural language text.

How GenAI Works – Data & Training

Content Generation:

- When receiving an input prompt, the model uses its learned knowledge to generate appropriate content.
- The output can be text, images (combined with diffusion models), audio, or video.
- The model doesn't just “remember” data—it can creatively synthesize and combine new information based on context

How GenAI Works – Data & Training

Fine-tuning and Reinforcement Learning

- After initial training, the model can be fine-tuned on domain-specific datasets to improve performance in targeted areas.
- The technique called Reinforcement Learning from Human Feedback (RLHF) helps the model improve based on user feedback, enhancing its usefulness and safety.

How GenAI Works – Data & Training

Real-world Applications

- GenAI is integrated into chatbots, automated content creation tools, code assistants, and artistic image generators.
- Businesses use GenAI to automate processes, boost efficiency, and reduce costs.

The Importance of Java/Spring in AI

Java is ideal for enterprise AI integration

Strengths:

- Reliability in finance, telecom, insurance
- Microservices-friendly
- Cross-platform, multithreaded
- Libraries for model embedding (ONNX, DJL)

Spring Boot:

- RESTful APIs, easy deployment
- Rich ecosystem (Security,...)

GenAI Integration Solution for Spring Boot

A Spring library for GenAI integration

Supported models:

- OpenAI (GPT-3.5, GPT-4)
- Azure OpenAI
- HuggingFace
- Mistral
- Ollama (local LLMs)

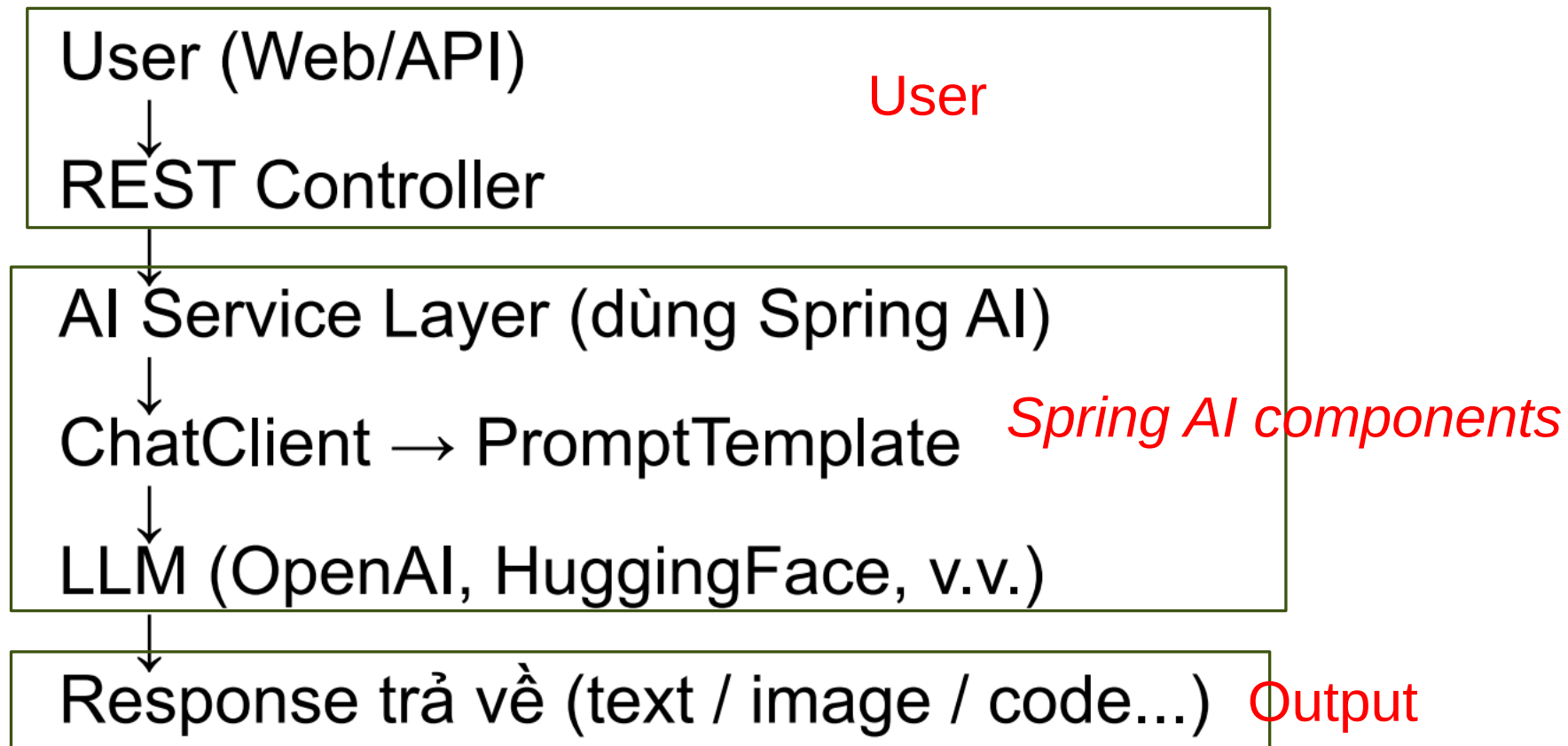


Key features:

- ChatClient
- PromptTemplate
- Memory
- VectorStore
- Retriever (for RAG)

Architectural Components and Processing Flow

GenAI Integration



Architectural Components and Processing Flow

RAG - Retrieval Augmented Generation

User question

User

↓
Embed câu hỏi thành vector

↓
VectorStore: tìm kiếm tài liệu tương tự (cosine similarity)

↓
Ghép kết quả vào prompt (contextual augmentation)

↓
LLM trả lời chính xác hơn

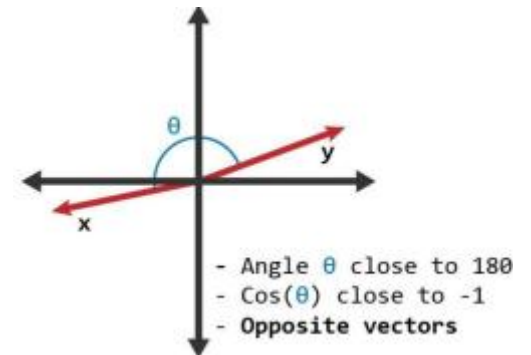
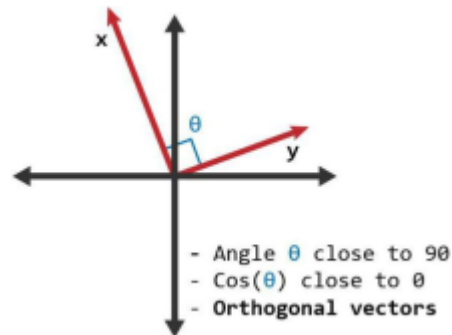
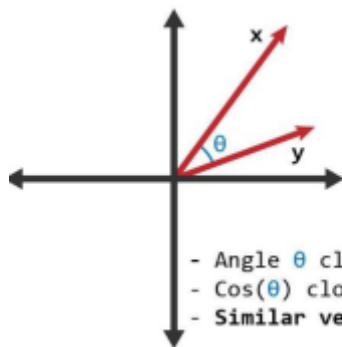
Cosine Similarity

Cosine Similarity is a technique in linear algebra used to measure the similarity between two vectors by calculating the angle between them in vector space.

Imagine two arrows pointing in space:

- If they point in the **same direction**, they're very similar.
- If they're **perpendicular**, they're unrelated.
- If they point in **opposite directions**, they're completely different

Cosine Similarity



Cosine Similarity

Cosine similarity ranges from **-1 to +1**:

+1 → vectors are identical in direction (maximum similarity)

0 → vectors are orthogonal (no similarity)

-1 → vectors are opposite (maximum dissimilarity)

In Natural Language Processing (NLP), convert words or sentences into **embedding vectors** — numerical representations of meaning

Cosine Similarity – Applications

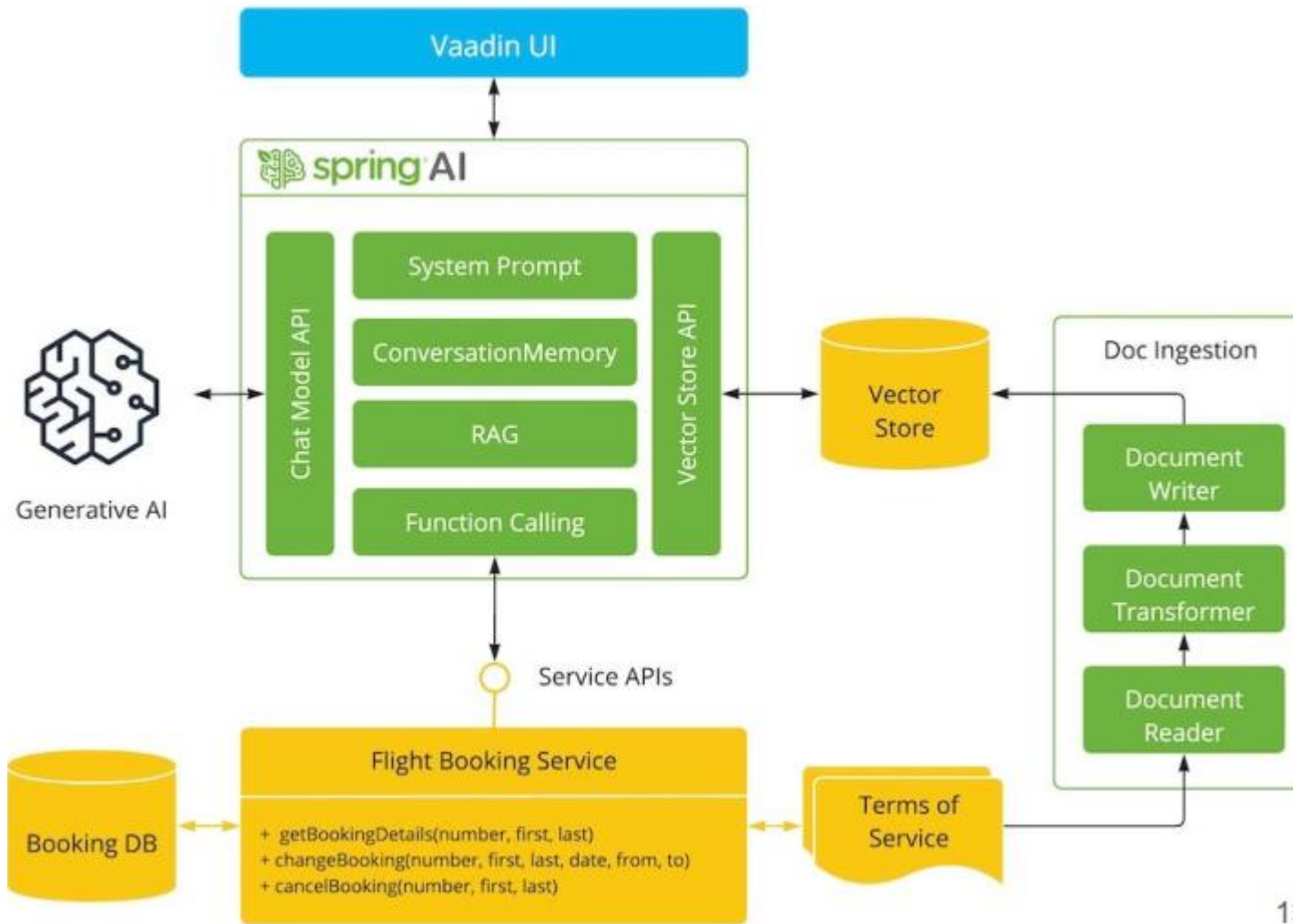
Vector Search: Find documents or text passages that are similar in content to a given query

RAG (Retrieval Augmented Generation): Retrieve the most relevant context to include in a prompt

Chatbots: Understand user intent even when different words are used

Recommendation Systems: Identify similar products based on vectorized descriptions

Semantic Search: Search by meaning rather than exact keywords



Lưu ý & Best practice

- ❑ **Bảo mật & cấu hình:**
 - API key nên lưu trong `.env`, `Vault`, hoặc CI/CD secrets
 - Không commit vào GitHub / GitLab
- ❑ **Tối ưu prompt:**
 - Tránh quá dài gây tốn token
 - Dùng template có cấu trúc rõ ràng
 - Chia context và câu hỏi riêng biệt
- ❑ **Logging & debugging:**
 - Ghi lại input, prompt, response để kiểm tra chất lượng
 - Log cả thời gian phản hồi (latency)
- ❑ **Modular hóa:**
 - Tạo interface cho AI service để dễ đổi model (OpenAI ↔ HuggingFace)
 - Test đơn vị nên mock response AI

Hướng phát triển:

- GenAI nội bộ: huấn luyện model hoặc fine-tune GPT
- GenAI + Workflow: AI tự động quyết định trong quy trình nghiệp vụ
- Tích hợp Agent: Dùng LangChain4j hoặc Spring AI Agent
- GenAI kết hợp dữ liệu thời gian thực: Kafka + RAG

Tài liệu tham khảo

- Spring AI docs: <https://docs.spring.io/spring-ai/reference/>
- Langchain4j: <https://github.com/langchain4j/langchain4j>
- OpenAI Platform:
<https://platform.openai.com/docs/overview>
- Qdrant: <https://qdrant.tech/>
- Weaviate Vector DB: <https://weaviate.io/>
- Ollama - Local LLMs: <https://ollama.com/>